

Electric Potential Energy

Goal.

- Build the idea of potential energy with gravity first: height above the ground, and what happens if you keep going *below* it.
- See the same idea for a charge pushed against a uniform field — for both a $+$ and a $-$ charge.
- Write the *full* (non-uniform) form of gravitational PE, and see that “higher” always means “more PE,” even though the value is negative.
- Use that same point-source structure to build the electric PE of two point charges, then a whole system of them.
- Combine two point charges’ PE to get the potential energy of a dipole in a field.

1. Gravitational PE: height, then a hole

A ball sits a height h above the ground. Ground level is our reference: $U = 0$ there. Draw the ball, the ground, and then let it fall *past* ground level into a hole of depth d — draw that too.

$$U(h) =$$

Now let the ball fall past ground level, into a hole of depth d .

$$U(\text{bottom}) =$$

$$\Delta U = U(\text{bottom}) - U(h) =$$

Check. The ball’s PE at the bottom of the hole came out negative. Is that a problem?

2. The same idea, for a charge in a uniform field

Replace gravity with a uniform electric field $\vec{E}$ pointing *down* (same visual sense as gravity), and try it with both signs of charge. Draw a $+q$ and a $-q$ in the field, below, each with an arrow for the force $q\vec{E}$ on it.

- Force on the $+q$: $q\vec{E}$ points _____ — that's its *natural* (free) direction of motion.
- Force on the $-q$: $q\vec{E}$ points _____.

Push each charge a distance d *against* its own natural direction — the same idea as raising the ball.

$$\Delta U_+ =$$

$$\Delta U_- =$$

- Both are _____ — pushing *against* a charge's natural direction always stores positive PE, regardless of the charge's sign.
- Let either charge go instead, and it moves along its *natural* direction: PE _____ either way — the field does the work.

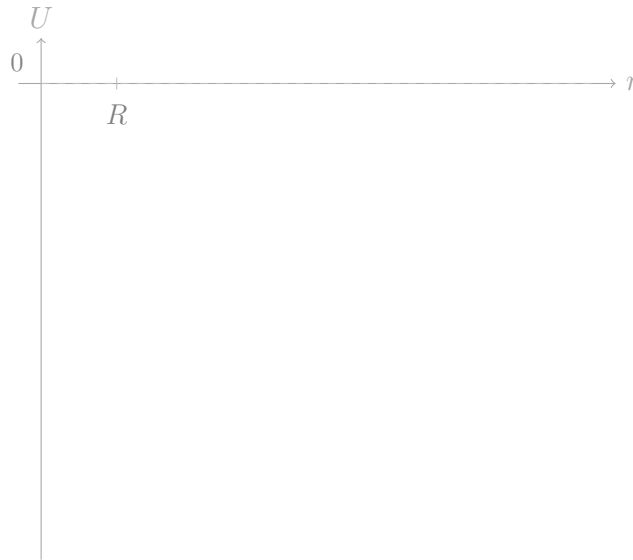
Check. A negative charge is released in this downward field. Which way does it move, and what happens to its PE?

3. Gravitational PE, the full (non-uniform) form

$U = mgh$ only works because g barely changes over any height you can reach near Earth's surface. Zoom out far enough to leave that “uniform field” approximation, and the real, general form (valid at any separation r between two masses) is:

$$U(r) =$$

Sketch U vs. r below.



- As $r \rightarrow \infty$, $U \rightarrow$ _____ — same rule as always: zero PE at infinite separation.
- Every value of $U(r)$ here is _____. But moving to a *larger* r (further from Earth, “higher”) always makes U _____. Don’t let the minus sign fool you: higher is *always* more PE, on this curve or any other.

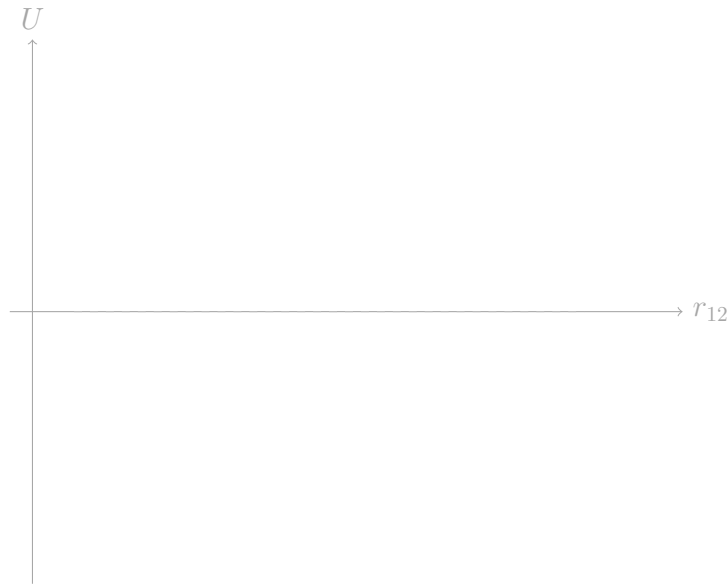
Check. True or false: since $U(r)$ is always negative here, moving *further* from Earth makes U smaller.

4. Electric PE of two point charges (from the class prep)

Pulling a pair of point charges apart, or pushing them together, takes work — work that gets stored as potential energy, exactly like the point-mass form above.

$$U(q_1, q_2) =$$

Sketch U vs. r_{12} below, for an opposite-sign pair and a like-sign pair, on the same axes. (Same shape you just drew for gravity — plus its mirror image.)



- As $r_{12} \rightarrow \infty$, $U \rightarrow$ _____.
- For three or more charges, sum over every _____ (not every charge).
- Opposite charges (+ and -): U is _____ at any finite separation, and $U \rightarrow 0$ only as $r \rightarrow \infty$. So $U = 0$ is actually the _____ value here — pulling the charges closer always finds a *lower* (more negative) PE.
- Like charges (same sign): U is _____ at any finite separation, and again $U \rightarrow 0$ as $r \rightarrow \infty$. Here $U = 0$ *is* the _____ value — you can't push like charges to a lower PE than infinitely far apart.

$U_{\text{total}} =$

Check. Two $+q$ charges are brought from far apart to a separation r . Is U positive or negative? What about a $+q$ and a $-q$?

5. The dipole in a field

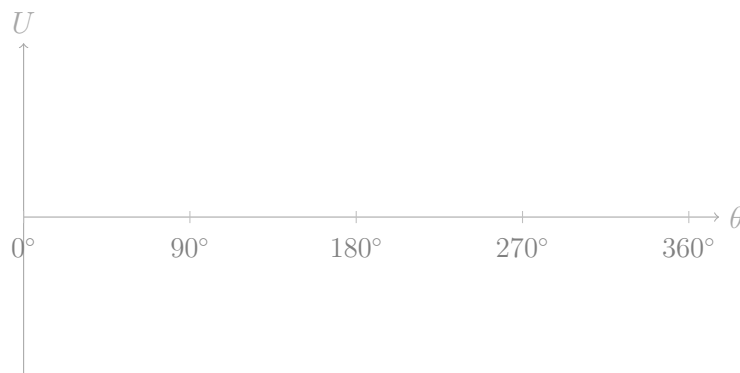
A dipole is a $+q$ and a $-q$ separated by $\vec{d}$ (dipole moment $\vec{p} = q\vec{d}$), sitting in a uniform field $\vec{E}$, tilted at angle θ between $\vec{p}$ and $\vec{E}$. Each charge sits at a different place along $\vec{E}$ — use §2's pushing idea on each charge, then add.

$$U(\theta) =$$

- At $\theta = 90^\circ$ (perpendicular to the field): $U =$ _____.
- At $\theta = 0^\circ$ (aligned with the field): $U =$ _____ — which is _____ than the $\theta = 90^\circ$ value!
- At $\theta = 180^\circ$ (anti-aligned): $U =$ _____.

Check. Which orientation is the most stable (lowest PE)? Is that the orientation where $U = 0$? What does this have in common with the ball in the hole?

6. Sketch it: U vs. θ



Check your curves, and the point-charge and uniform-field results above, against the *Assembling Charges*, *PE vs. Separation*, and *Electric Dipole in a Field* sims — open their **Info** panels.